Software Engineering Documentation

Project: Blood Glucose Management Webapp

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Chapter 1

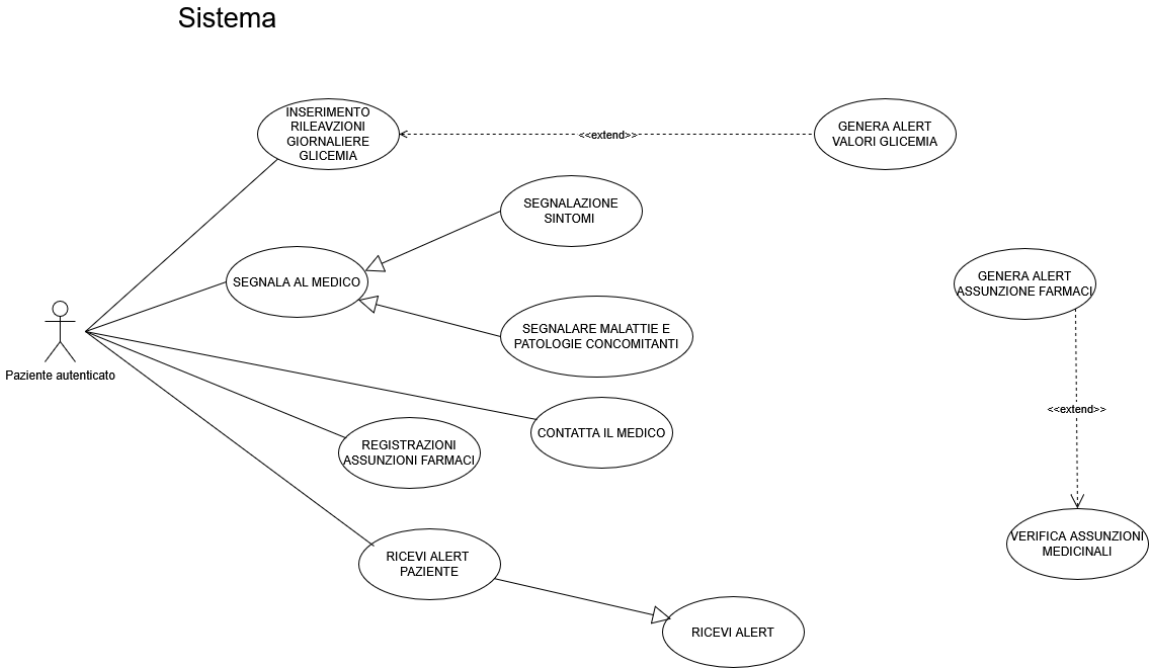
User Actions

The developed Webapp serves three types of users:

- Patient
- Doctor
- Admin

All user categories must authenticate to access their reserved area. We will now look at the use cases and their sequence diagrams. For clarity in visualizing the images, we will divide the use cases by user, although they are interconnected.

1.1 Patient Use Cases



After authenticating, the patient is redirected to their homepage, where they will find:

- A chart showing their blood glucose trends over time
- A button to insert blood glucose readings
- The latest recorded blood glucose entries
- A form for communicating symptoms and diseases
- An area to mark taken medications
- A form to contact the doctor

We now list the use cases shown in the image below:

- Insert blood glucose reading
- Report pathologies, diseases, and/or symptoms to the doctor
- Daily recording of taken medications
- Contact the doctor

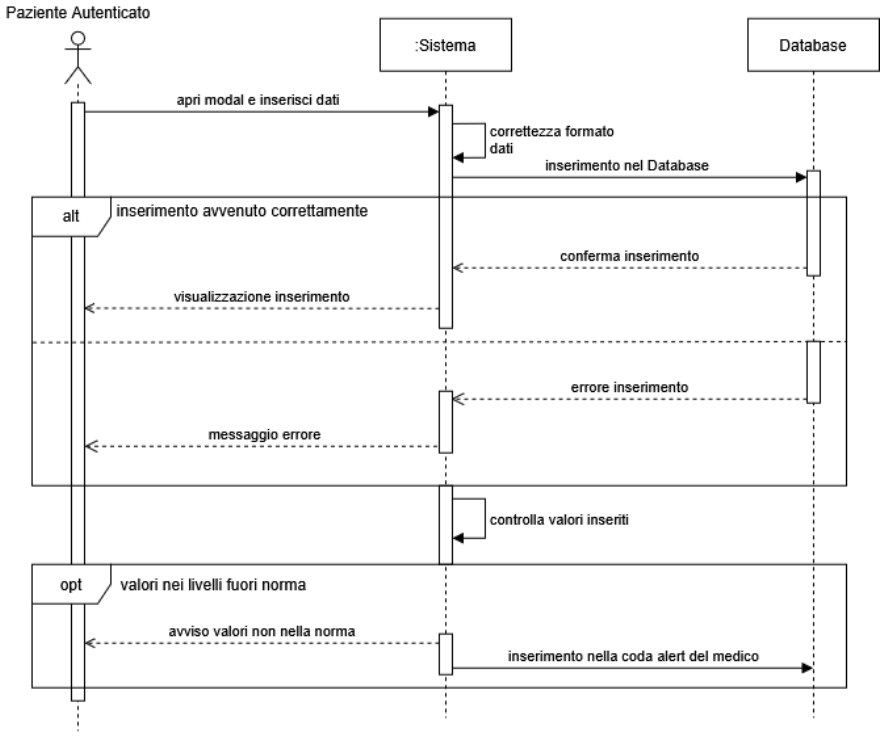
1.1.1 Insert Blood Glucose Reading

The patient must be able to insert their glycemic level. By clicking the "insert glucose" button, a modal appears where they can input their glycemic level. Specifically, when a patient inserts a reading that falls outside the normal range (80-130 mg/dL and maximum 180mg/dL after meals), the system records the reading and generates an alert to inform the doctor.

Actors: Patient
Preconditions: The patient must be authenticated
Steps:

1. The patient clicks the "New reading" button
2. The patient fills the glucose modal field that appears
3. The patient clicks the "send" button
4. The validity of the entered data is checked
5. Alerts for Doctor and Patient may be created in case of out-of-range values

Postconditions: The glycemic data is inserted into the database



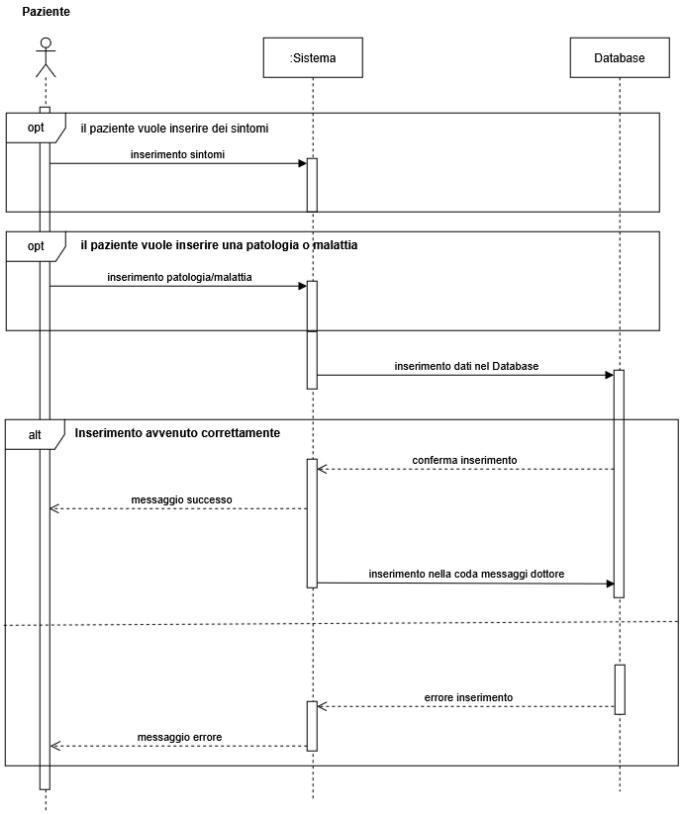
1.1.2 Reporting Pathologies, Diseases, and/or Symptoms

The patient can report any pathologies, diseases, and/or symptoms to their doctor. In the dedicated box, the patient can select various symptoms from checkboxes or type them freely in the text field along with any diseases and pathologies.

Actors: Patient
Preconditions: The patient must be authenticated
Steps:

1. The patient selects the checkboxes for the experienced symptoms
2. The patient fills in the text field with pathologies and diseases
3. The patient clicks the "send report" button
4. The report is communicated to the doctor

Postconditions: The report is inserted into the database and notified to the doctor



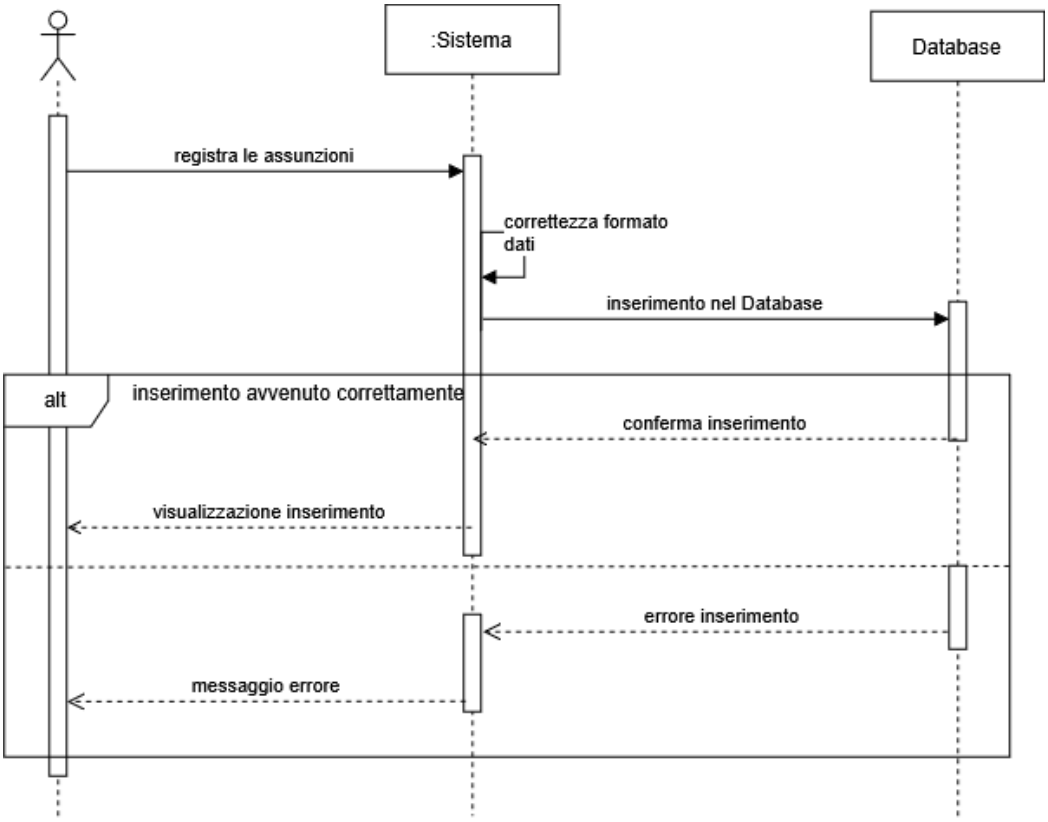
1.1.3 Daily Recording of Taken Medications

The patient can record their daily taken medications. In the appropriate box, there will be the list of medications the patient should take during the day; they simply need to select the taken ones and press the "send report to doctor" button so they can be recorded.

Actors: Patient
Preconditions: The patient must be authenticated
Steps:

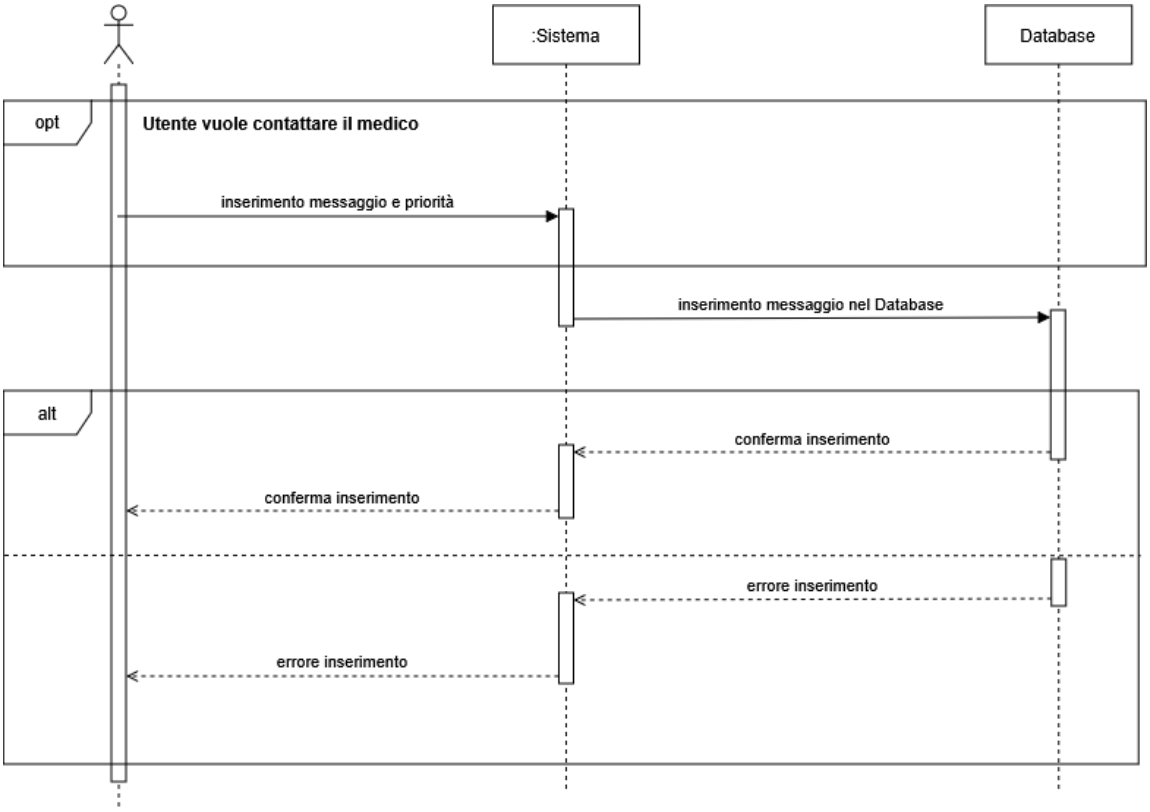
1. The patient selects the taken medications from the list
2. The patient clicks the "send report to doctor" button
3. The entered data is then checked
4. Registration confirmation is provided

Postconditions: The taken medications are recorded in the database

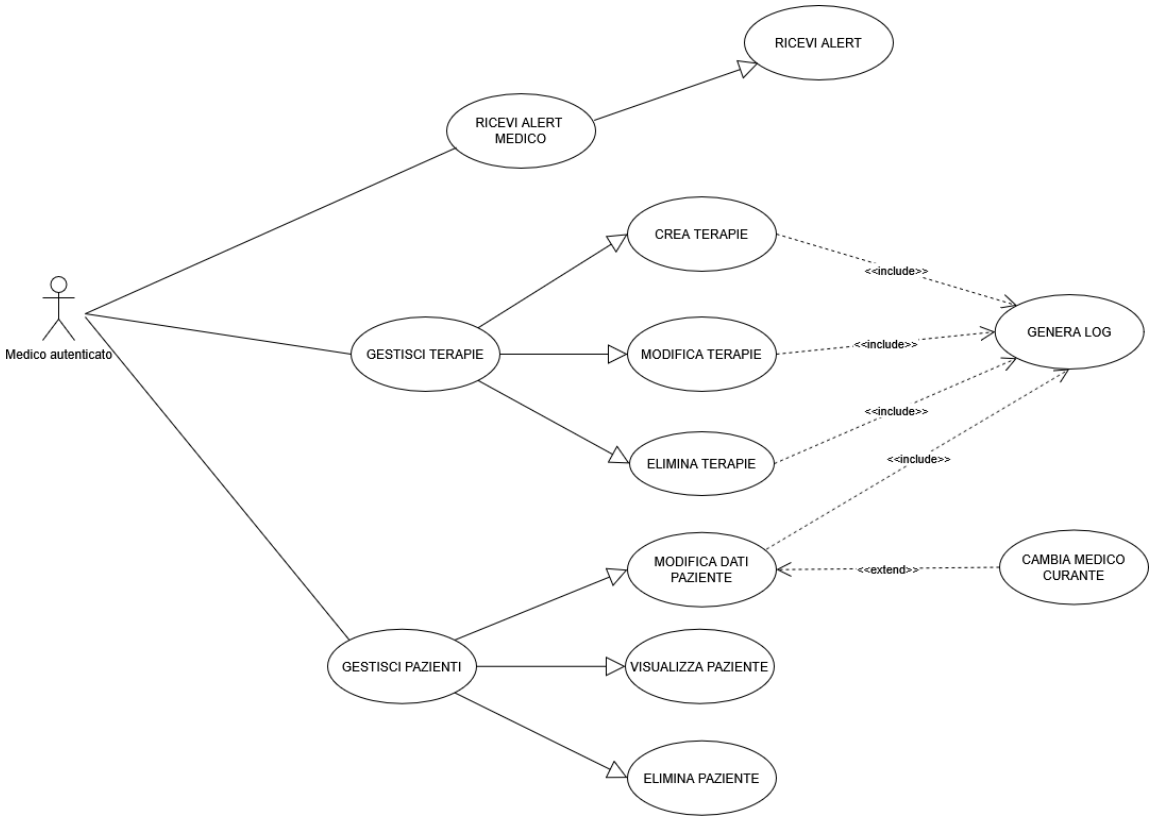


1.1.4 Contact the Doctor

The Patient can contact their doctor using a dedicated form. In the appropriate box, the patient must fill out the text field and select the message priority among low, medium, and high; then press "send" to send the message.



1.2 Doctor Use Cases

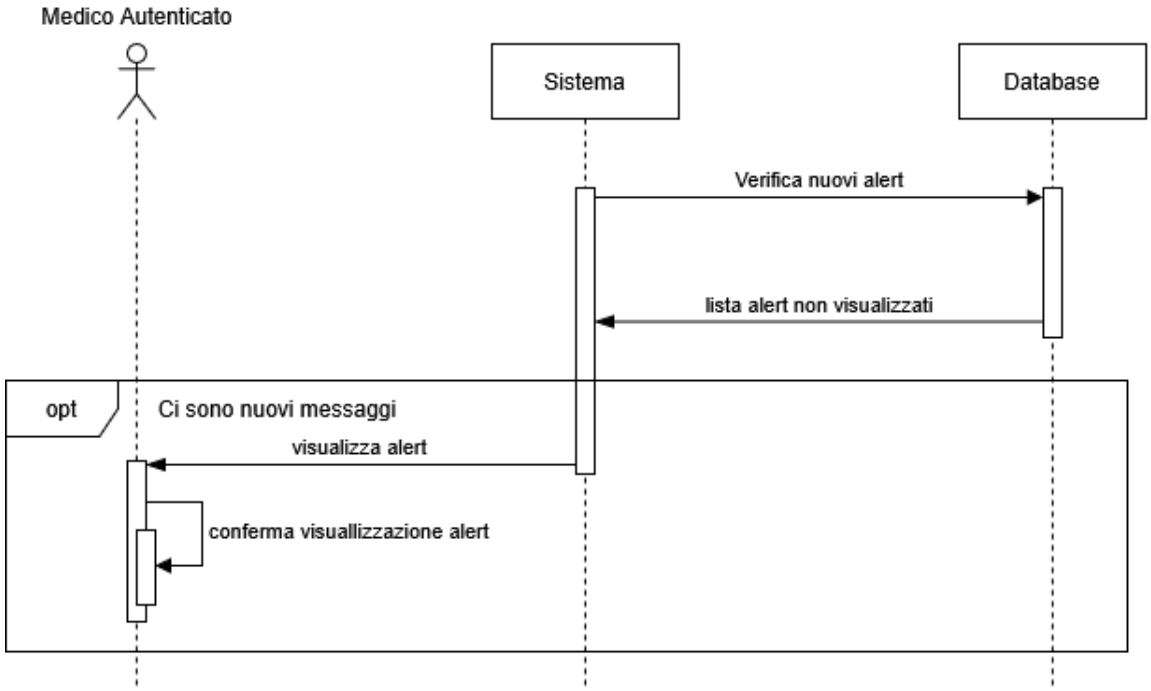


After authenticating, the doctor is redirected to their homepage, where they will find:

- A box with the list of received messages and alerts
- An area for managing patient therapies
- An area for managing patients

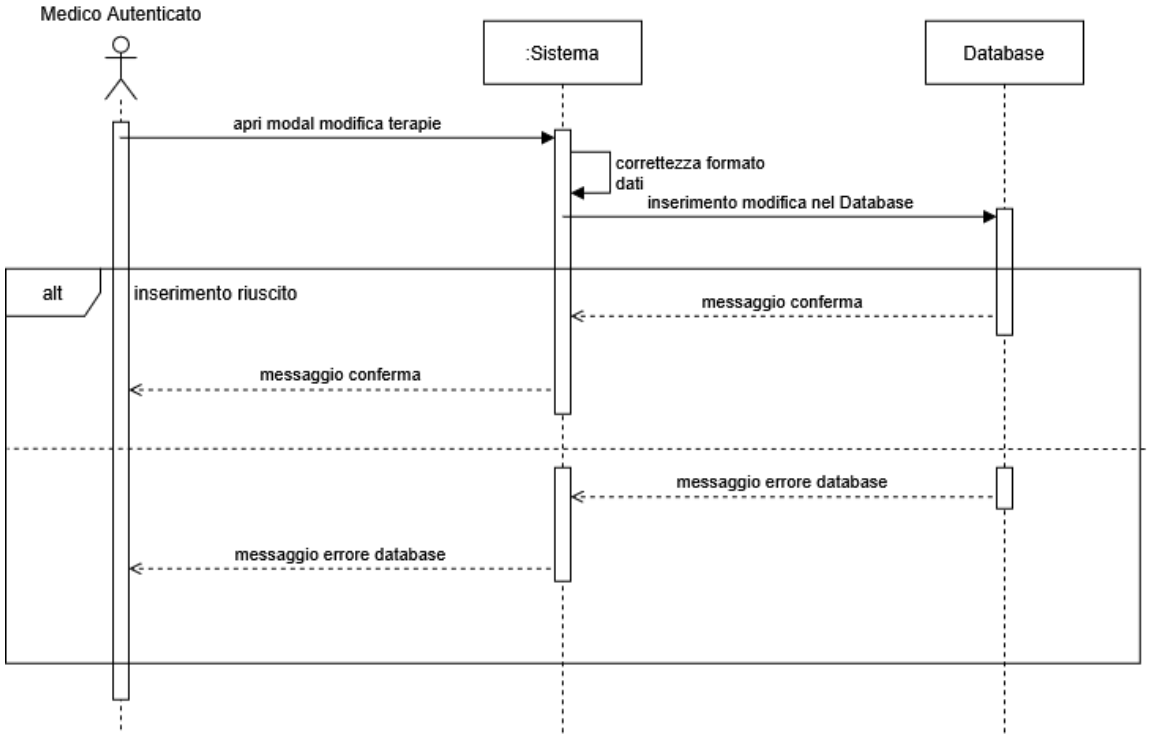
1.2.1 View Alerts for the Doctor

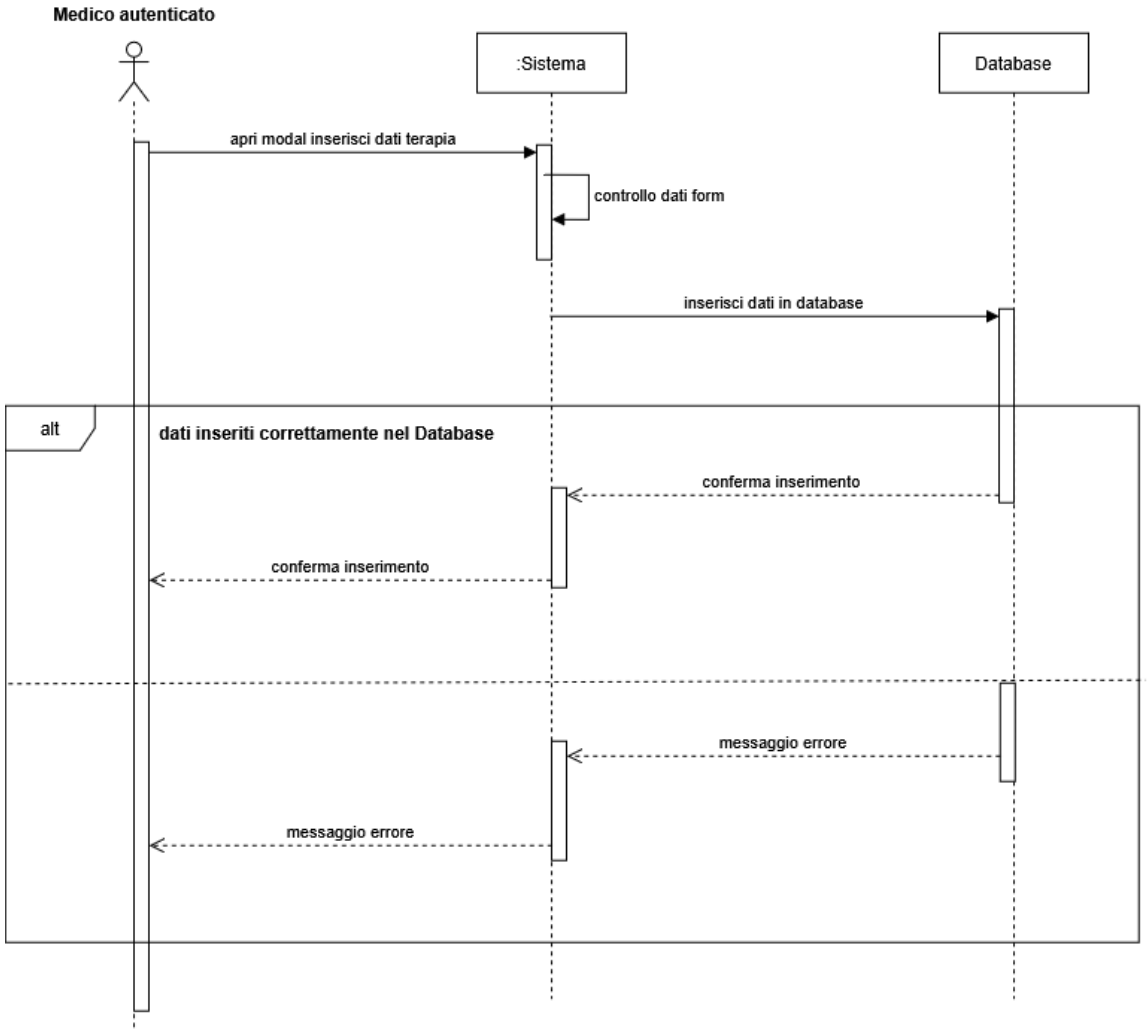
The doctor will have the list of messages and alerts on the right side of their homepage; when they have read the alert, they can press "read" to mark it and remove it.

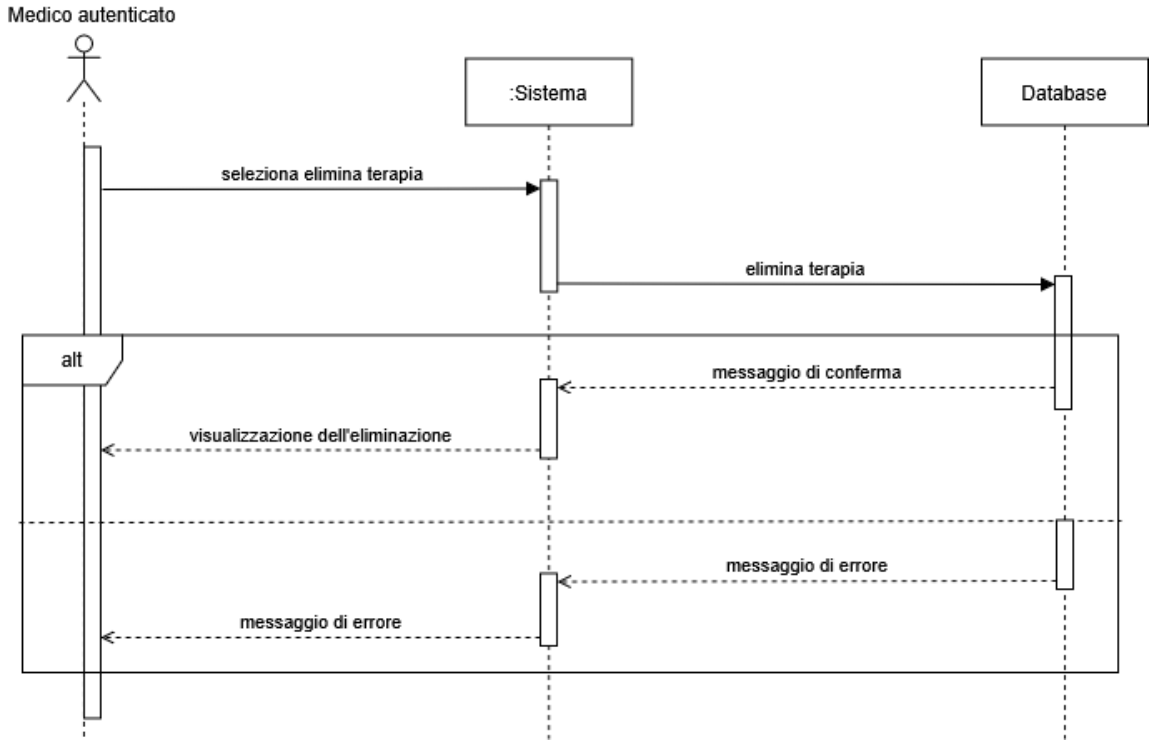


1.2.2 Therapy Management

The doctor can modify a therapy by clicking the pencil button next to it, delete it using the trash can button, or create a new one using the "New Therapy" button.

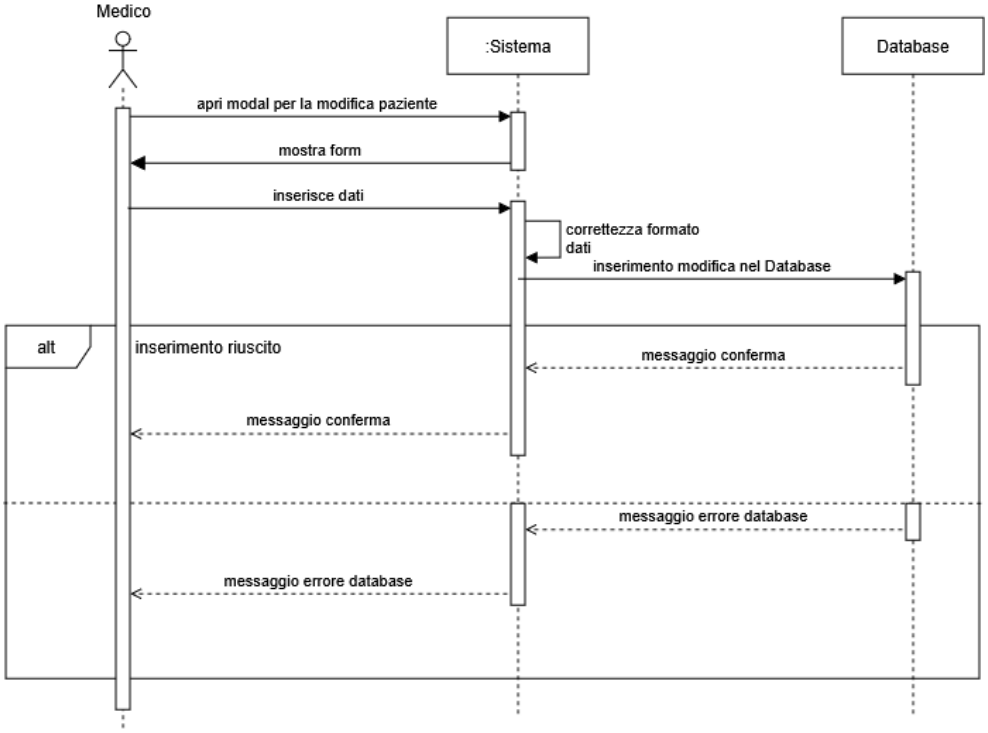

Medico Autenticato
:Sistema
Database
apri modal modifica terapie
correttezza formato dati
inserimento modifica nel Database
alt
inserimento riuscito
messaggio conferma
messaggio conferma
messaggio errore database
messaggio errore database

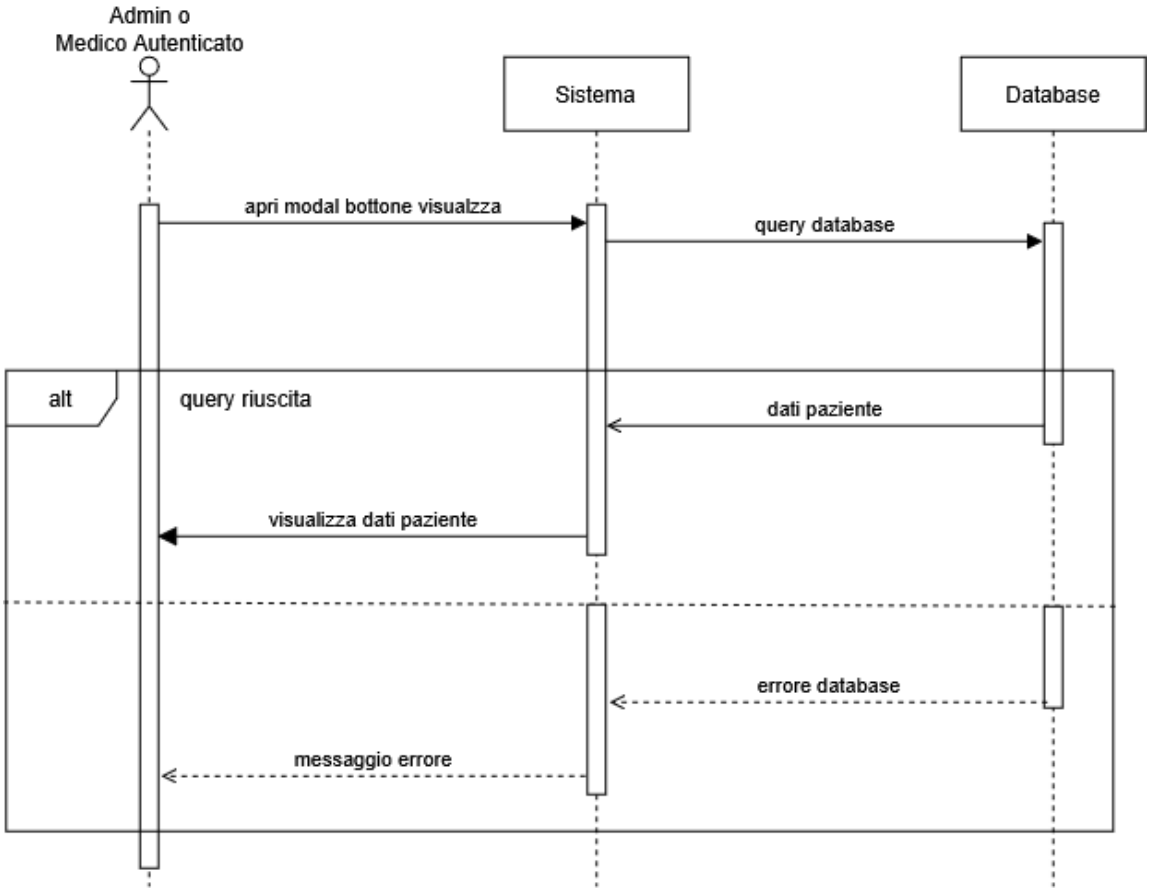

Medico autenticato
:Sistema
Database
apri modal inserisci dati terapia
controllo dati form
inserisci dati in database
alt
dati inseriti correttamente nel Database
conferma inserimento
conferma inserimento
messaggio errore
messaggio errore

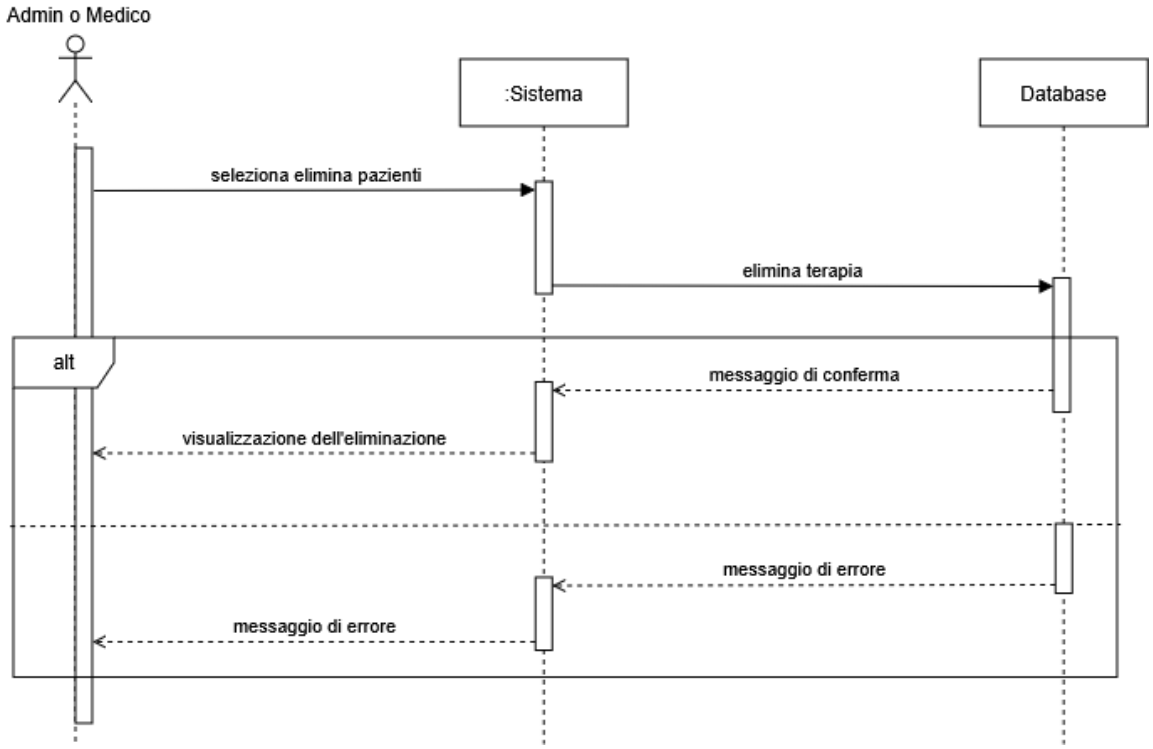


1.2.3 Patient Management

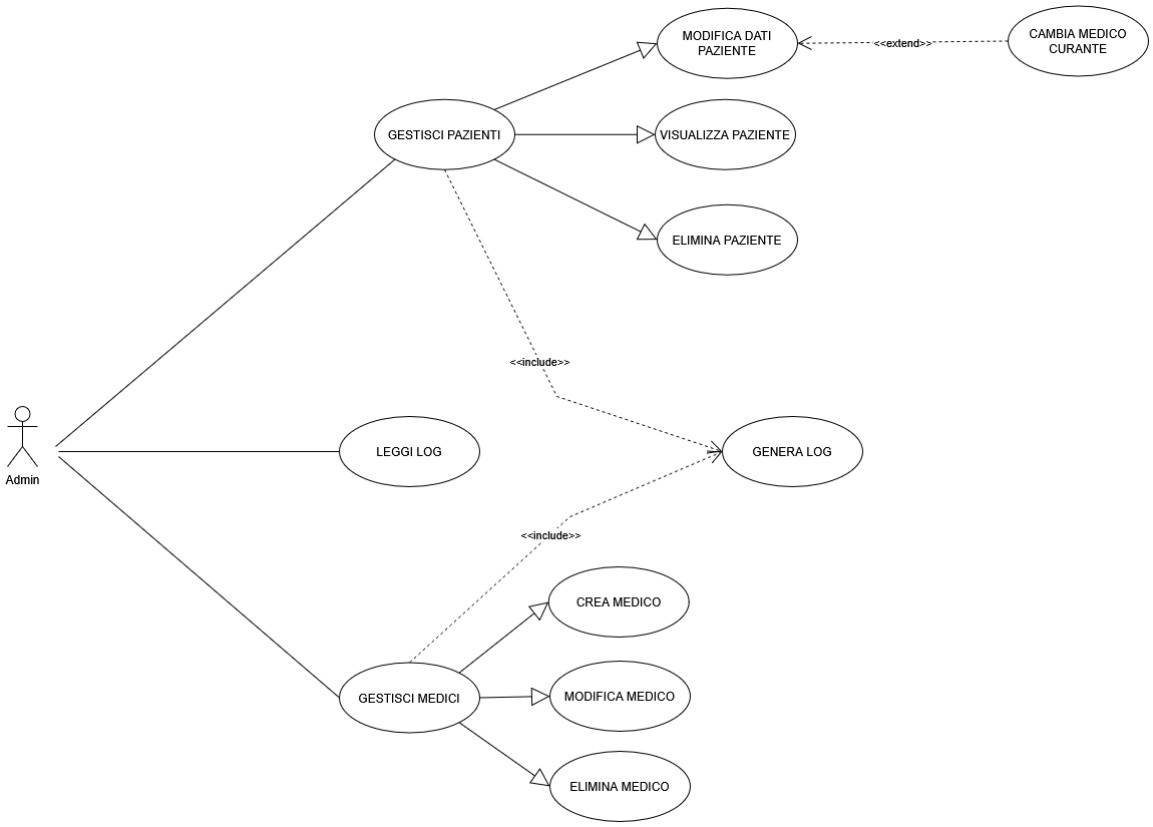
The doctor can decide to edit a Patient's information, delete it, or view the glycemic trend details using the respective icons.


Medico
:Sistema
Database
apri modal per la modifica paziente
mostra form
inserisce dati
correttezza formato dati
inserimento modifica nel Database
alt
inserimento riuscito
messaggio conferma
messaggio conferma
messaggio errore database
messaggio errore database

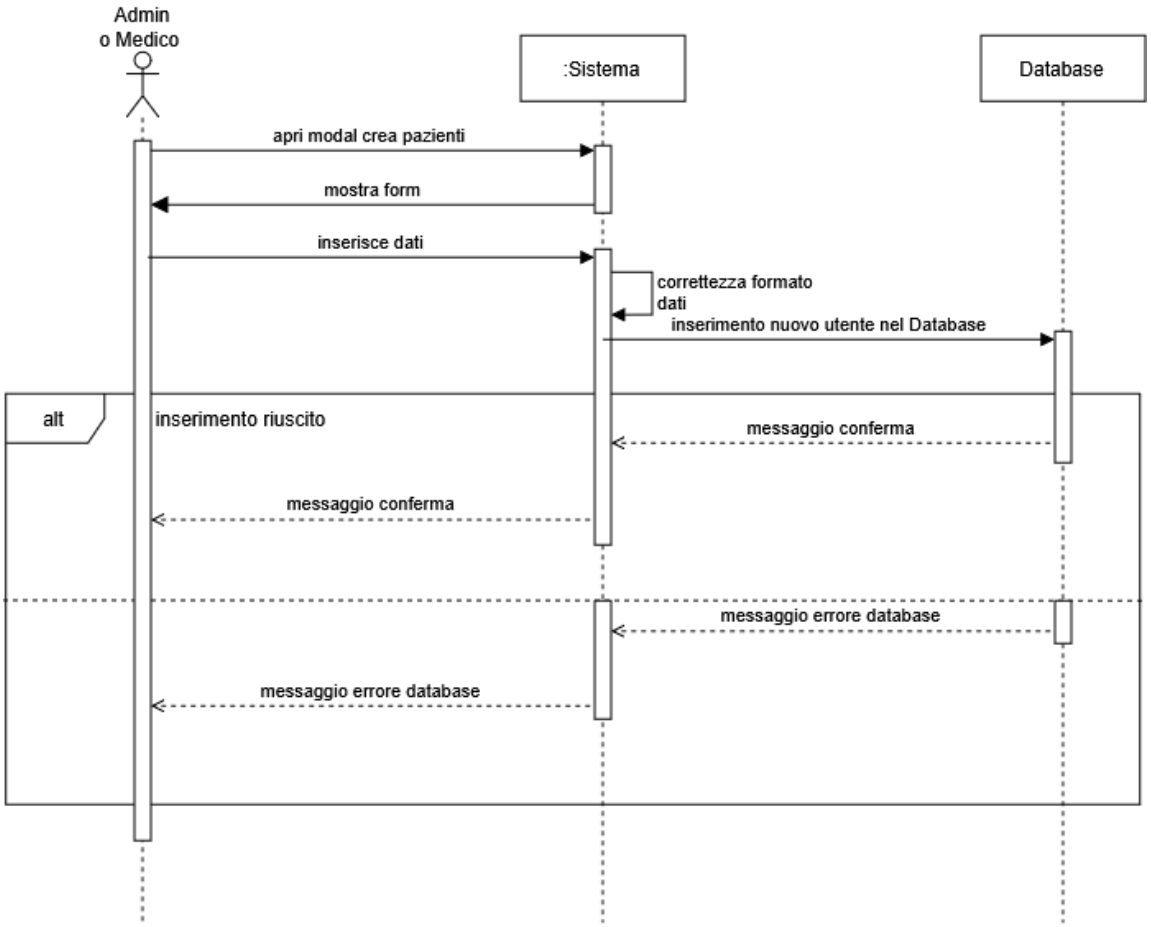

Admin o
Medico Autenticato
Sistema
Database
apri modal bottone visualzza
query database
alt
query riuscita
dati paziente
visualizza dati paziente
errore database
messaggio errore

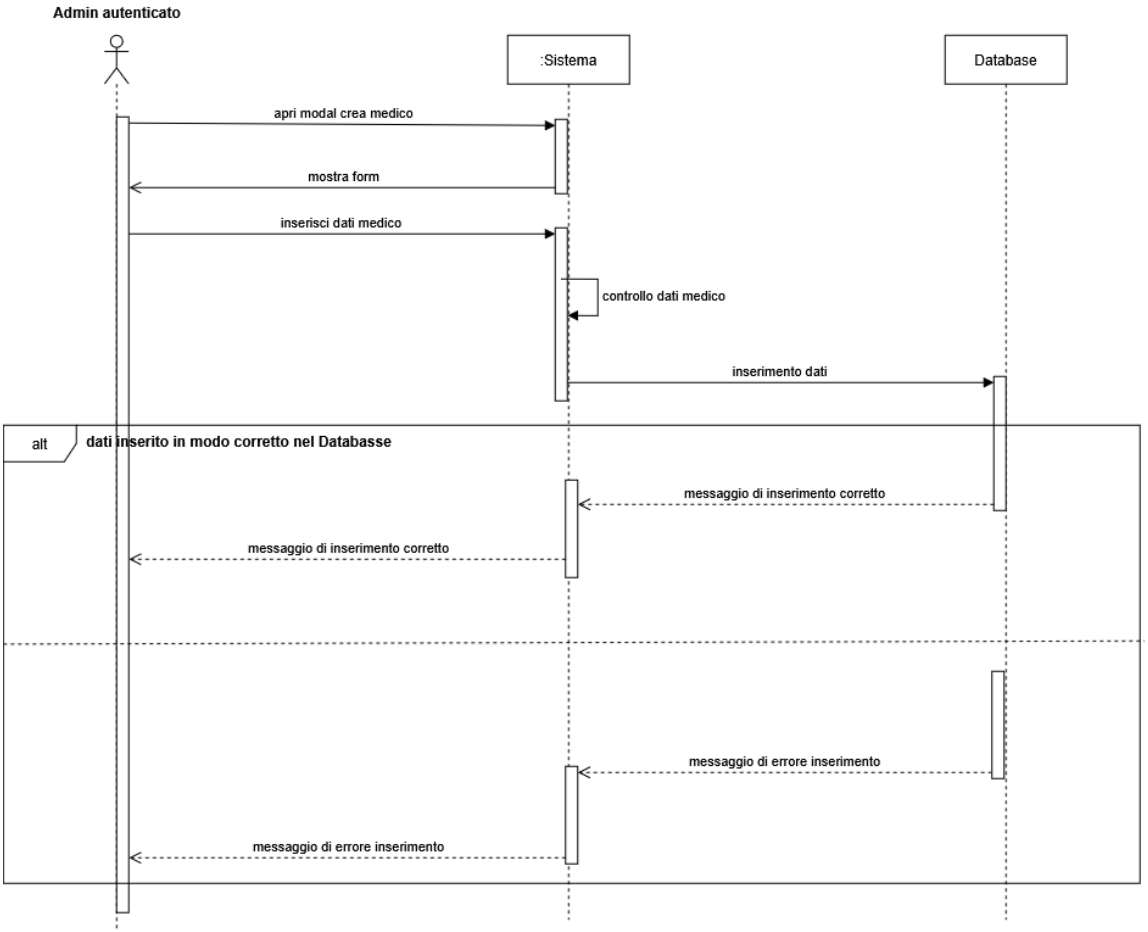

Admin o Medico
:Sistema
Database
seleziona elimina pazienti
elimina terapia
alt
messaggio di conferma
visualizzazione dell'eliminazione
messaggio di errore
messaggio di errore

1.3 Admin Use Cases



After authenticating, the admin is redirected to their homepage, where they will find features to manage users (Doctors and Patients) and view system logs.

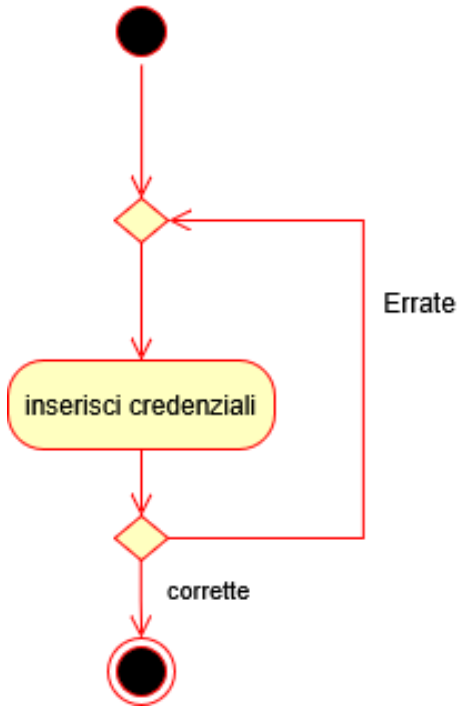

Admin
o Medico
:Sistema
Database
apri modal crea pazienti
mostra form
inserisce dati
correttezza formato
dati
inserimento nuovo utente nel Database
alt
inserimento riuscito
messaggio conferma
messaggio conferma
messaggio errore database
messaggio errore database


Admin autenticato
:Sistema
Database
apri modal crea medico
mostra form
inserisci dati medico
controllo dati medico
inserimento dati
alt
dati inserito in modo corretto nel Databasse
messaggio di inserimento corretto
messaggio di inserimento corretto
messaggio di errore inserimento
messaggio di errore inserimento

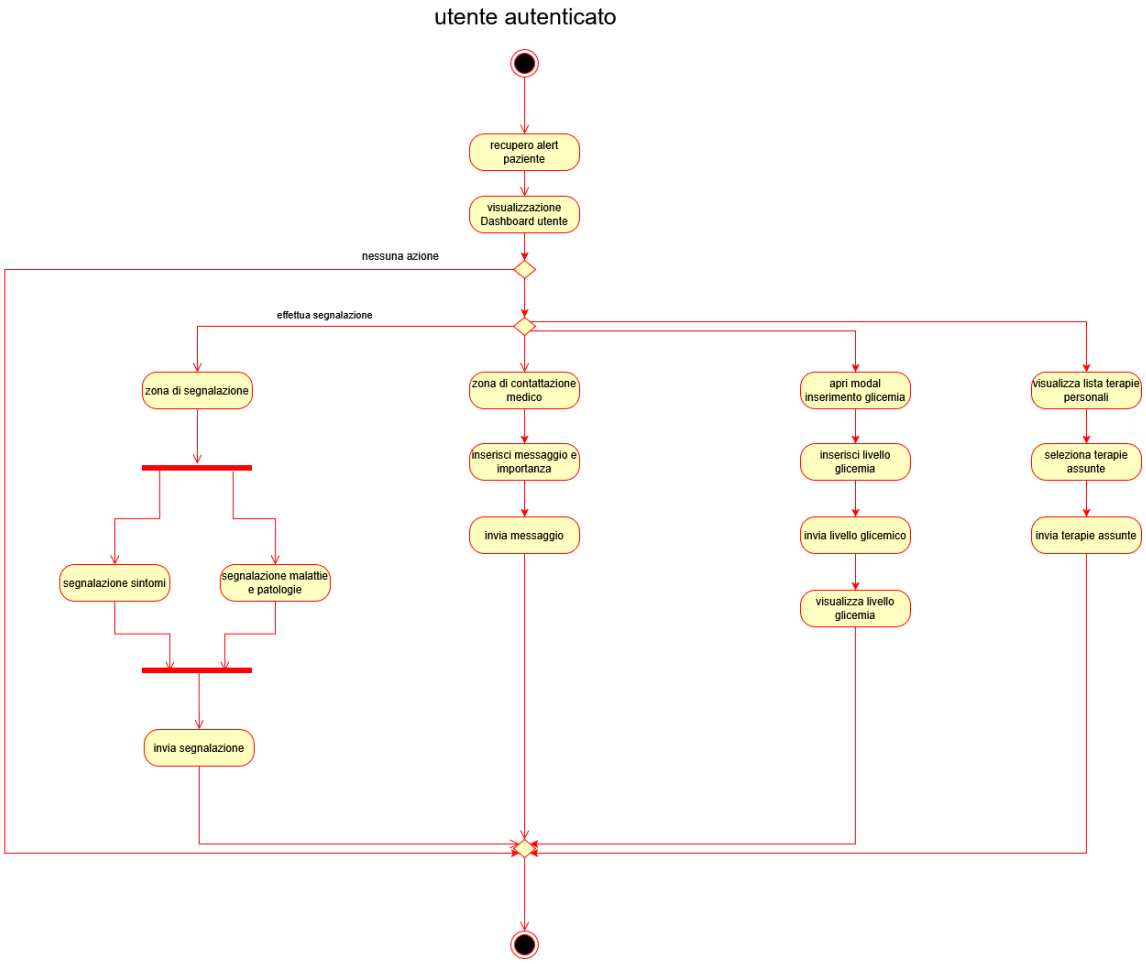
Chapter 2

Activity Diagrams

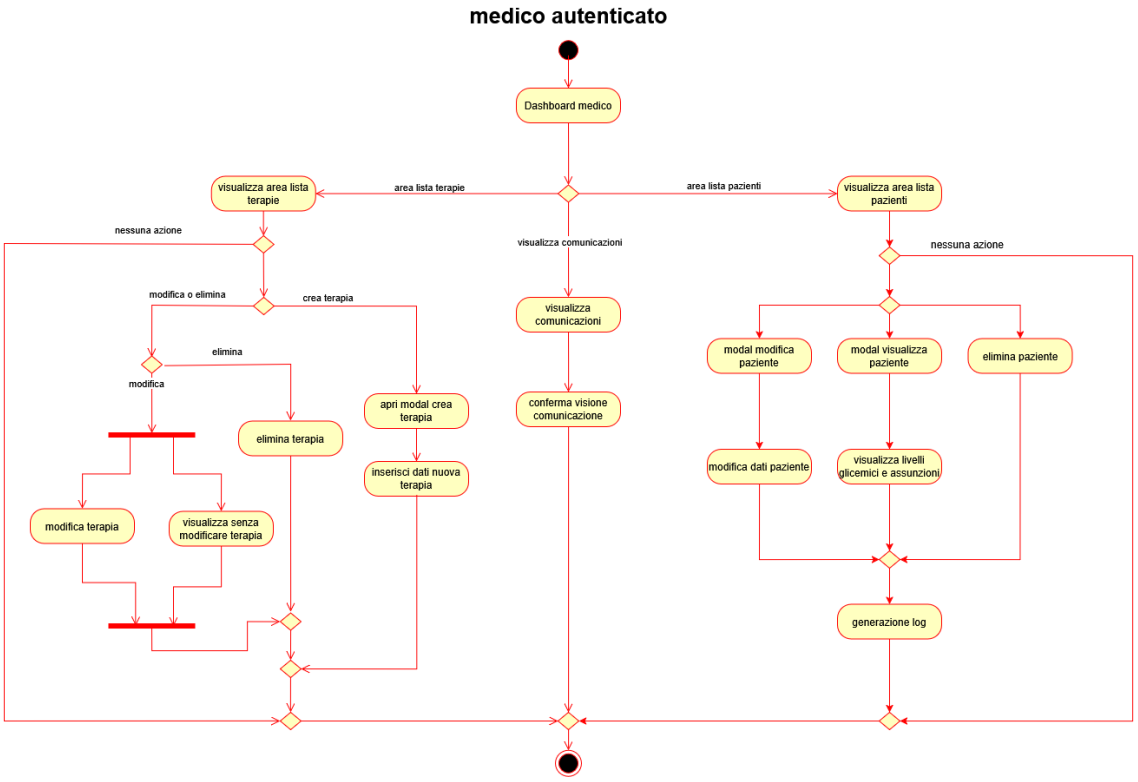
2.1 Authentication Diagram



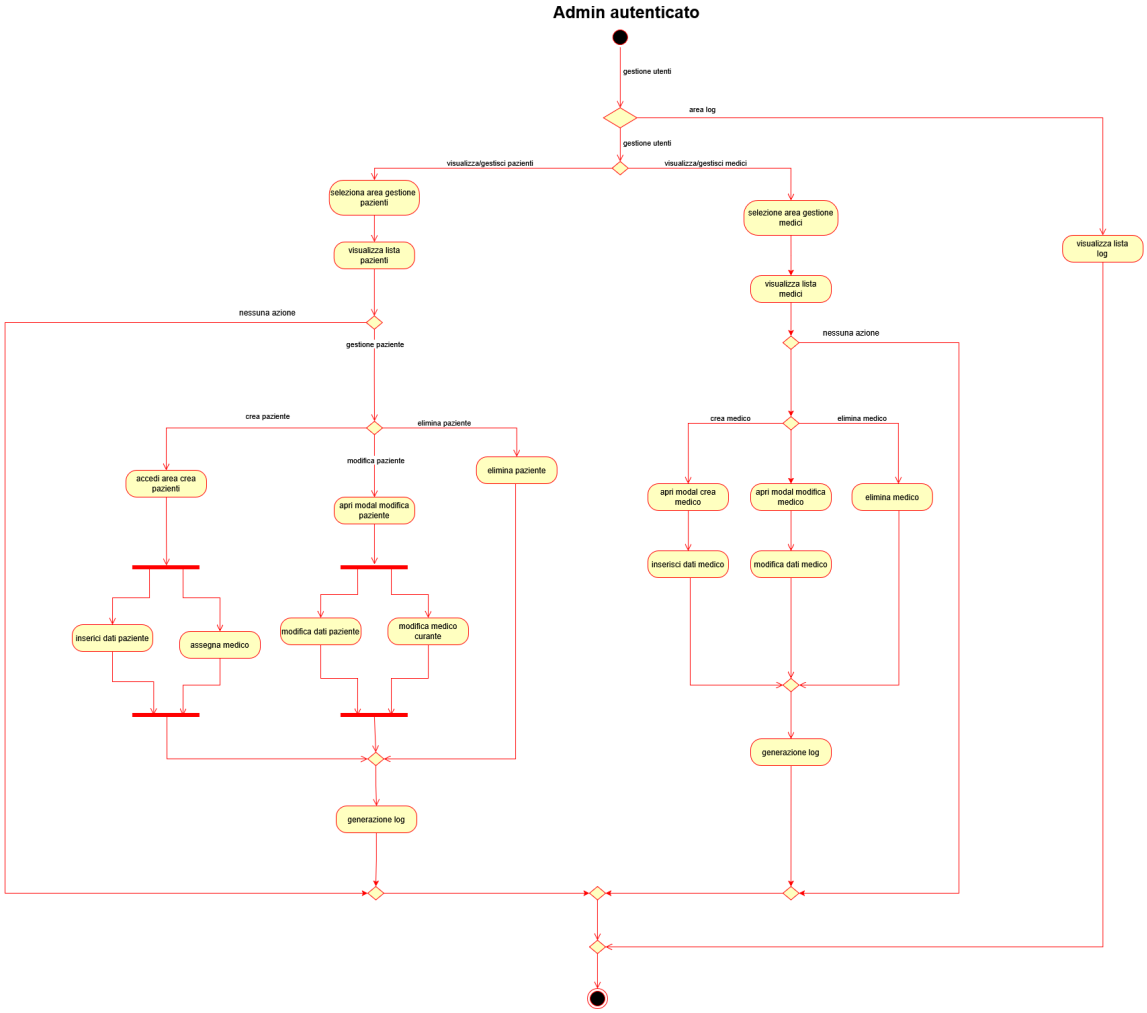
2.2 Patient Diagram



2.3 Doctor Diagram



2.4 Admin Diagram



Chapter 3

UML Diagrams

3.1 Class Diagram - Entity Package

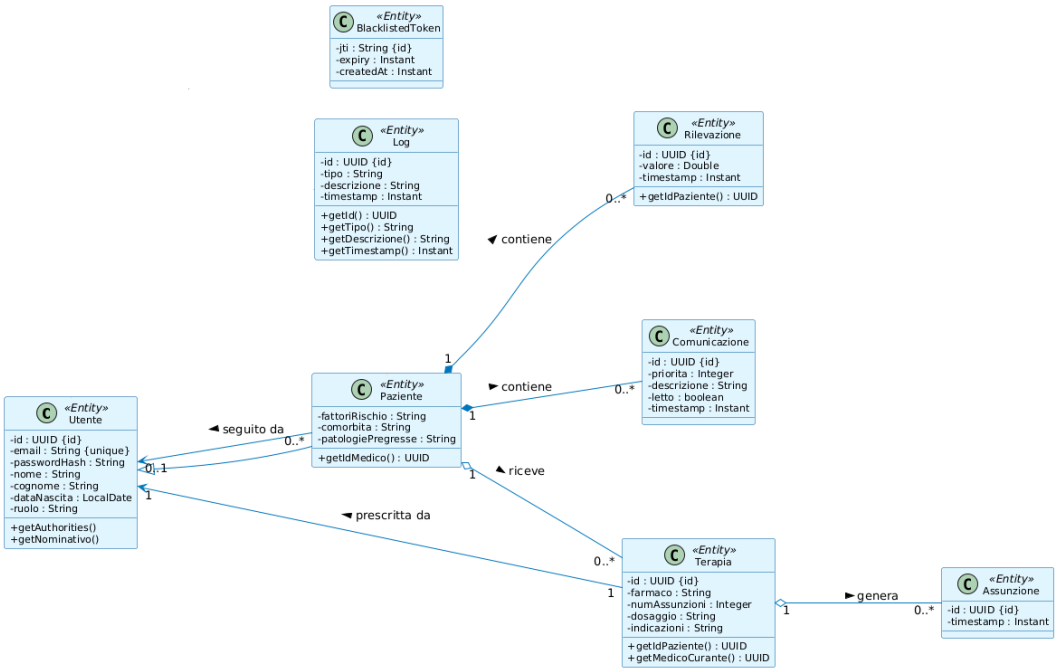
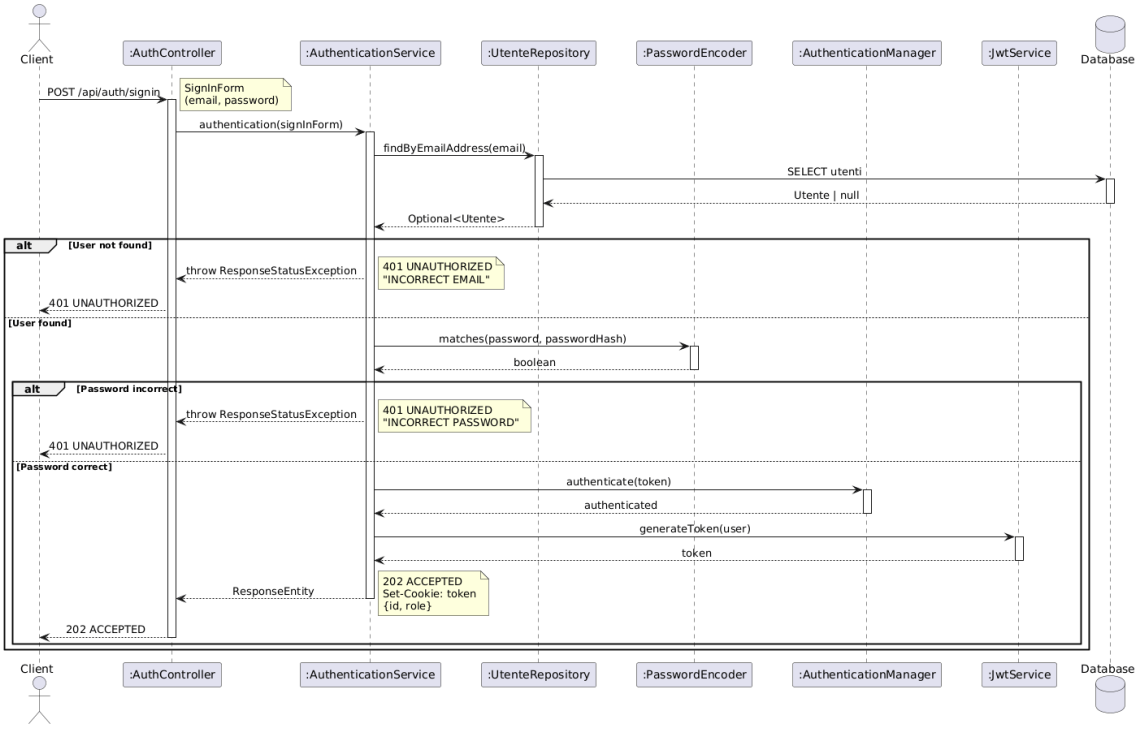


Figure 3.1: Class Diagram - Entity Package

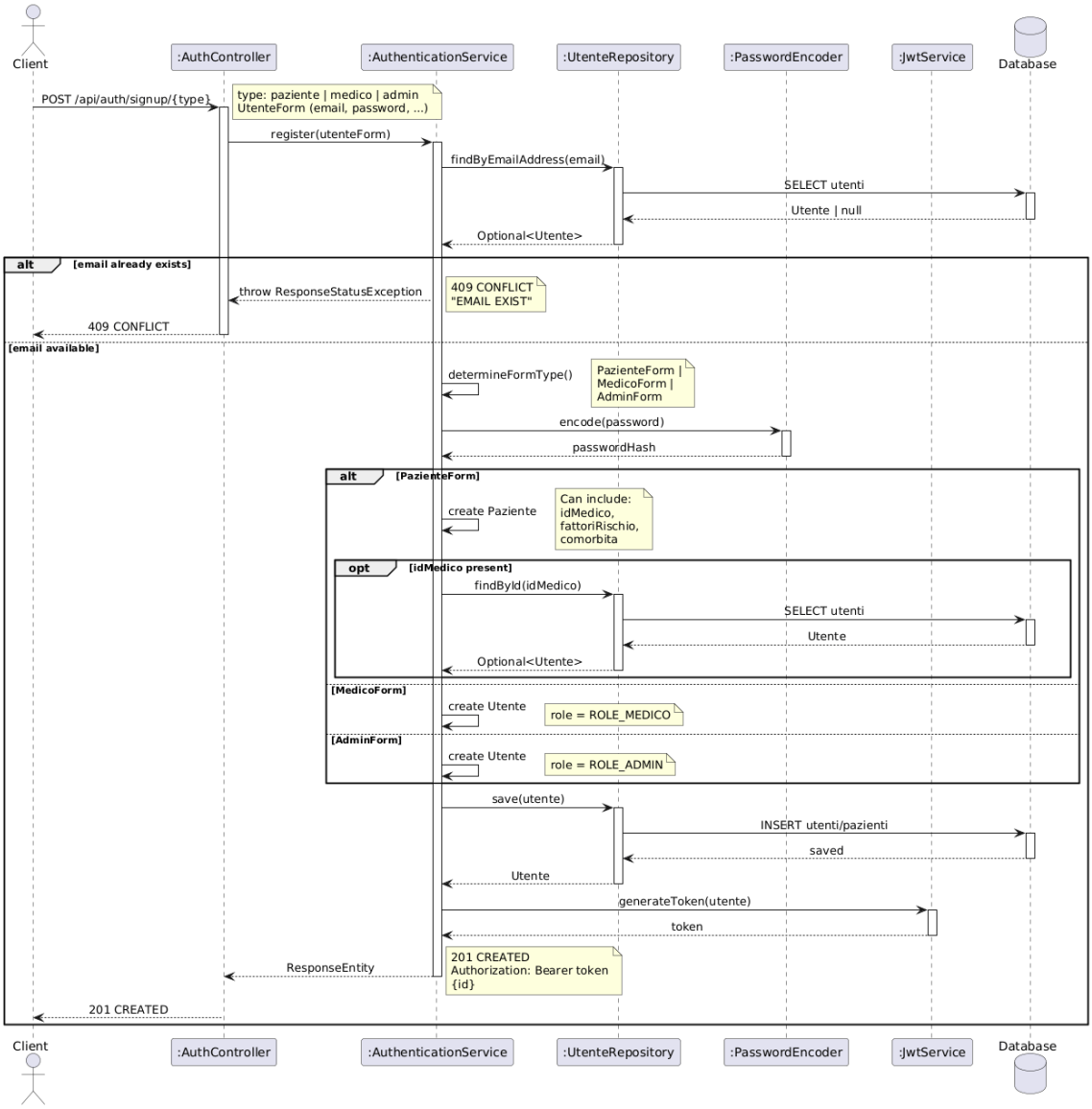
The `entity` package defines the object representation of the data schema, including entities like Utente, Paziente, Terapia, Assunzione, Rilevazione, Comunicazione, Log, and BlacklistedToken.

3.2 UML Sequence Diagrams

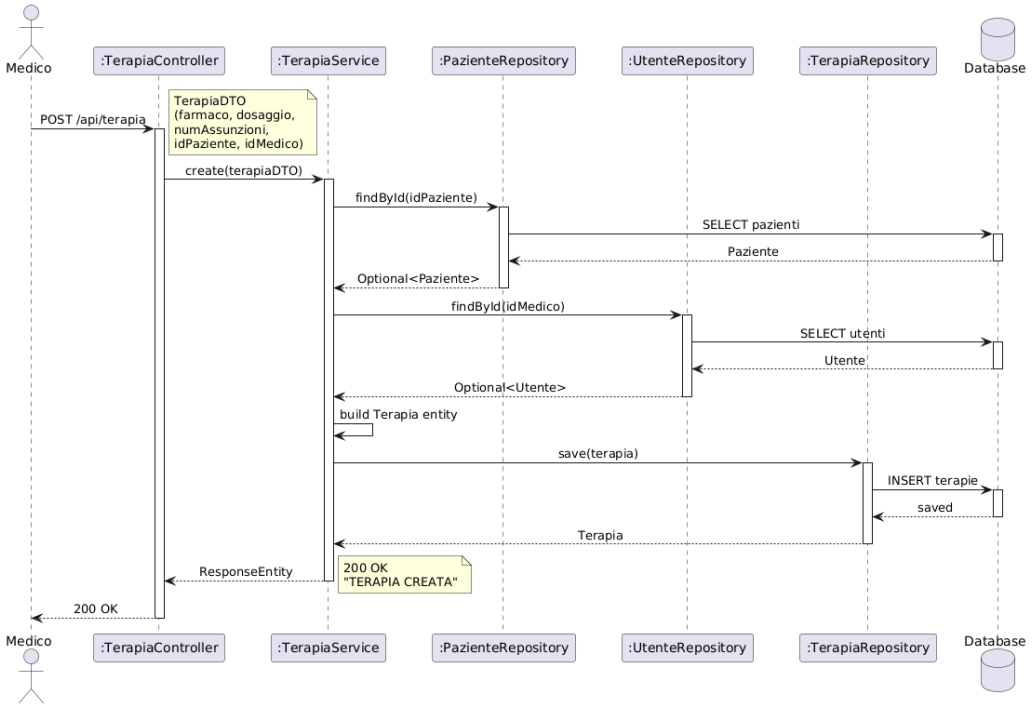
3.2.1 Authentication (Sign-In)



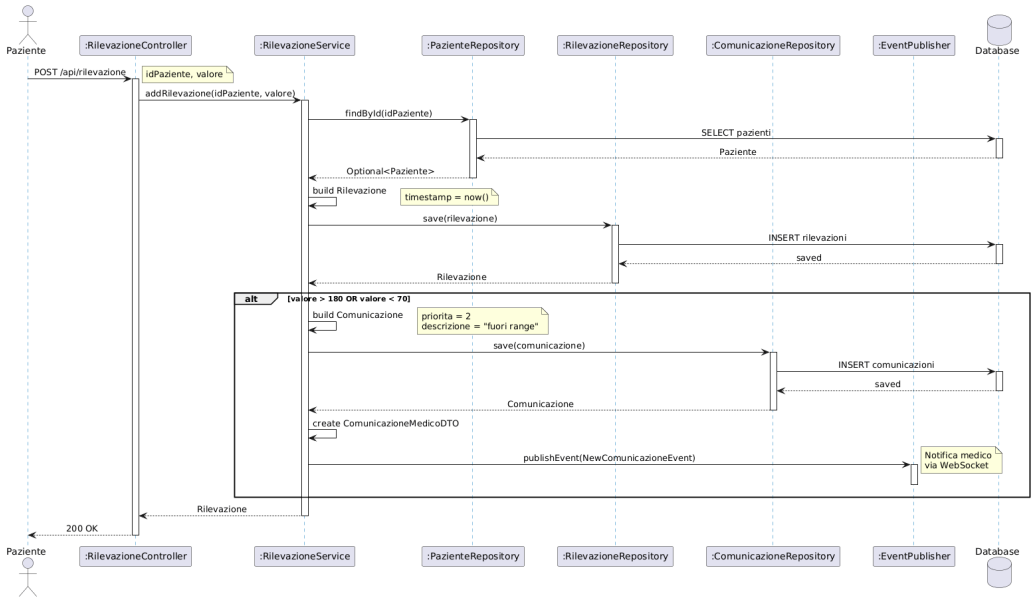
3.2.2 User Registration



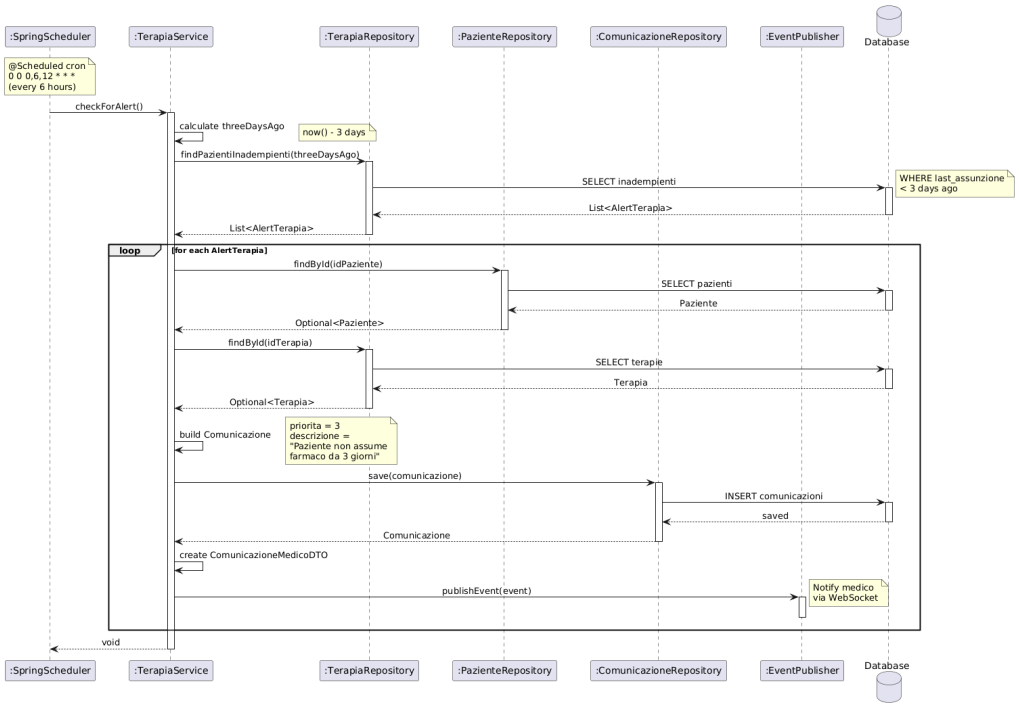
3.2.3 Therapy Creation



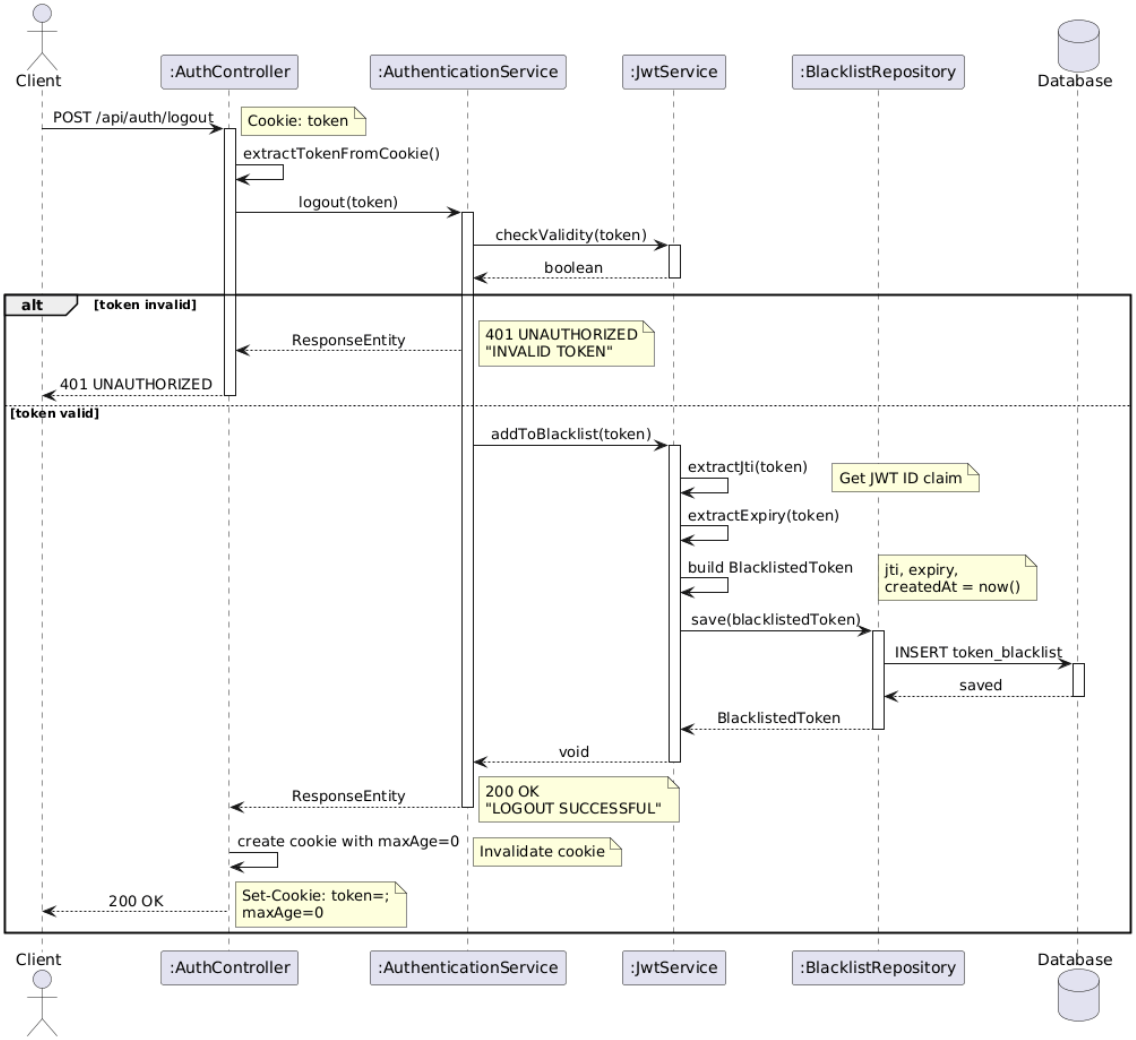
3.2.4 Add Reading with Automatic Alert



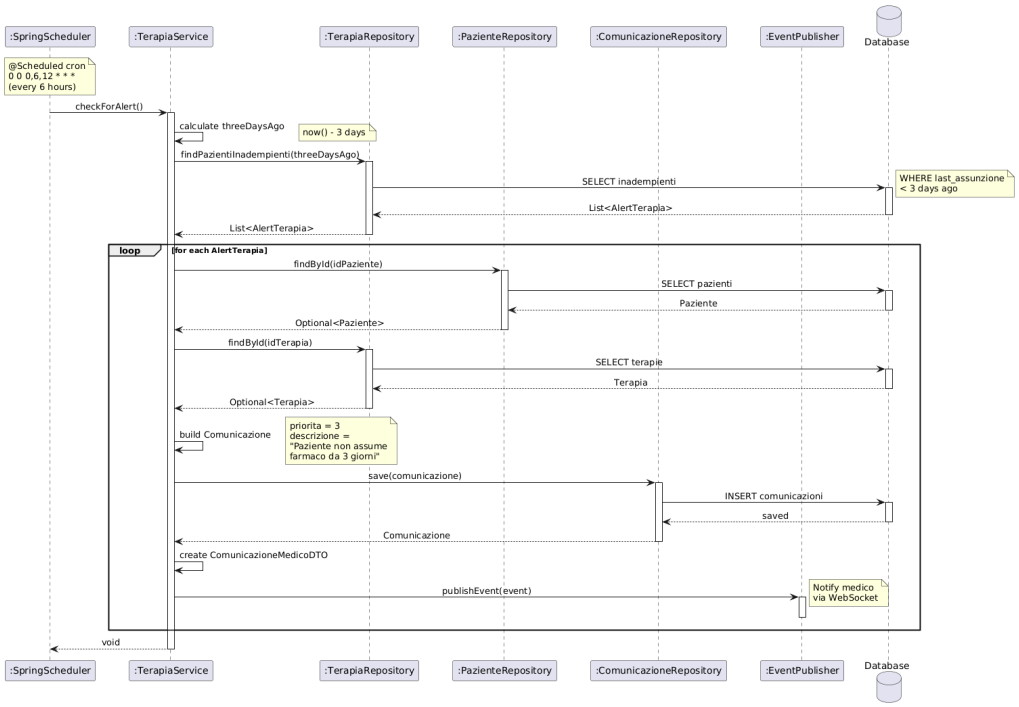
3.2.5 JWT Filter for Request Authentication



3.2.6 Logout


Client
:AuthController
:AuthenticationService
:JwtService
:BlacklistRepository
Database
POST /api/auth/logout
Cookie: token
extractTokenFromCookie()
logout(token)
checkValidity(token)
boolean
alt
[token invalid]
ResponseEntity
401 UNAUTHORIZED
"INVALID TOKEN"
401 UNAUTHORIZED
[token valid]
addToBlacklist(token)
extractJti(token)
Get JWT ID claim
extractExpiry(token)
build BlacklistedToken
jti, expiry,
createdAt = now()
save(blacklistedToken)
INSERT token_blacklist
saved
BlacklistedToken
void
ResponseEntity
200 OK
"LOGOUT SUCCESSFUL"
create cookie with maxAge=0
Invalidate cookie
200 OK
Set-Cookie: token=;
maxAge=0

3.2.7 Scheduled Automatic Therapy Check

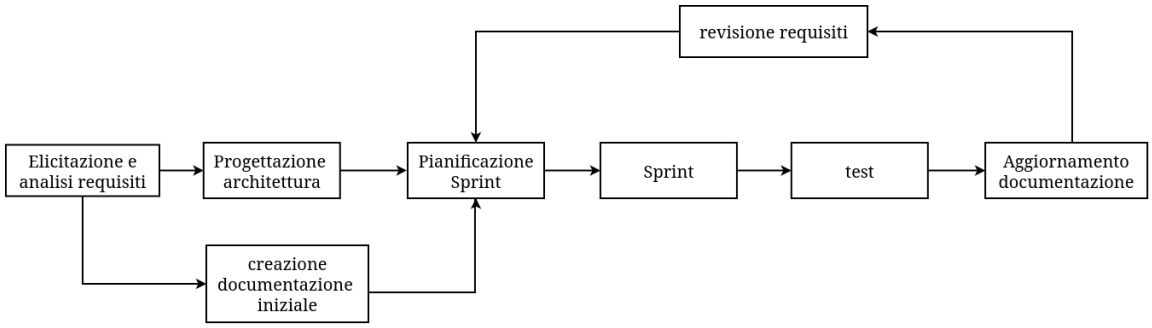


Chapter 4

Software Development

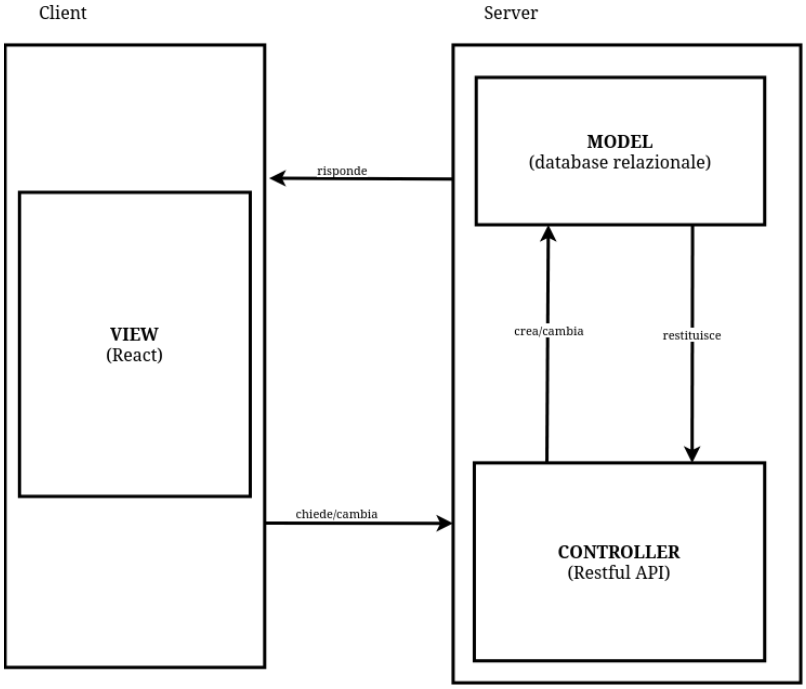
4.1 Development Methodology

An iterative and incremental Agile-based process model was adopted for the system development, characterized by high operational flexibility depending on project specificities. The process includes Requirement Analysis, Sprint planning, Implementation, Testing, and Documentation updates.



4.2 Architecture

The software adopts a *Client-Server* architecture with elements of the *Model-View-Controller* pattern.



4.3 Implementation

Backend (Spring Boot) is structured into Entity, Repository, Service, Controller, DTO, Config, Filter, and Exception packages. The Frontend is developed with React 18 and TypeScript.

Chapter 5

Testing

Automated testing was performed using JUnit and Mockito to verify the correct functioning of services. Manual tests and external user tests were also conducted to ensure usability and correct operation of the system.